



Functional Requirements Specification (FRS)

TASCO Insurance × VETC Growth Platform — Business & Product

iorta TechNXT

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References such as "doc 04" point to the companion documents listed in the Business and Product Documentation index (00-index.pdf).

0. Conventions

Item	Convention
ID	FR- <i>nnn</i> . IDs are stable and referenced by docs 04 (user stories) and 05 (traceability).
Priority	MoSCoW: M = Must for MVP, S = Should (MVP if capacity allows, else Phase 2), C = Could (Phase 2+), W = Won't in this release (recorded for scope clarity).
Track	T1 AI voice bot · T2 Lead scoring and enrichment · T3 Renewal engine · T4 Value beyond discount · NB New-business scope extension requested by the client · PL Platform, governance and compliance.
Rule kind	Business configuration that governs the behaviour. Rule kinds live in <code>config/rules/<kind>.json</code> and are changed only through the maker-checker workflow (FR-082 – FR-088). <code>rbac</code> (roles, separation-of-duties pairs, restricted rule kinds) lives in <code>config/security/rbac.json</code> and is changed through code review.
Status	Build status as of 07 October 2026 (codebase at <code>/src</code> , verified by the automated test suite). Built = implemented and tested; a note may name a planned production integration · Partial = implemented with a functional gap noted · Planned = not yet built (roadmap).
Actors	Roles as in <code>config/security/rbac.json</code> : <code>admin</code> , <code>executive</code> , <code>campaign_manager</code> , <code>telesales_agent</code> , <code>telesales_supervisor</code> , <code>rule_author</code> , <code>rule_approver</code> , <code>compliance_officer</code> , <code>data_steward</code> , <code>claims_handler</code> , <code>partner_manager</code> , <code>auditor</code> , <code>support_engineer</code> , <code>customer</code> , <code>partner_api</code> .

Capability map

#	Capability	FR range
A	Data and master data management (MDM)	FR-001 – FR-011
B	Lead intelligence	FR-012 – FR-019
C	Journeys and messaging	FR-020 – FR-030
D	AI voice bot	FR-031 – FR-039
E	Telesales	FR-040 – FR-044
F	Sales and issuance	FR-045 – FR-052
G	Products and rating	FR-053 – FR-057
H	Value and benefits	FR-058 – FR-062
I	Partners and fleet	FR-063 – FR-070
J	Claims FNOL	FR-071 – FR-073
K	Customer self-service and privacy	FR-074 – FR-081
L	Rules governance	FR-082 – FR-088
M	Identity and access	FR-089 – FR-096

#	Capability	FR range
N	Audit	FR-097 – FR-099
O	Operations, reporting and user interfaces	FR-100 – FR-110

A. Data and master data management (MDM)

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-001	Batch ingestion. The system shall accept batches of up to 5,000 source records from an authorised user (data:ingest) or partner channel. Each batch is tagged with a batch ID, source name and lineage entry. Only allow-listed fields are kept: recordId, source, plateRaw, phoneRaw, fullName, tollClass, seatsDeclared, usageDeclared, ownerType, tagActivatedAt, policy, lastInspectionDate, declaredExpiry, partnerId, firstRegisteredYear.	recordId is mandatory. Unknown fields are dropped. PII fields are encrypted at rest.	M	T2	Built
FR-002	Licence plate normalisation. The system shall normalise Vietnamese plates from dirty formats (for example 30a-123.45, 30A 12345) to a canonical key (30A12345) and display form (30A-123.45), and derive the province from the 2-digit code. Invalid plates are rejected and raise a DQ issue invalid_plate.	Format: 2-digit province + series (1–2 letters, or letter + digit for motorbikes, e.g. 59X1-123.45) + 4–5 digits; province must be known.	M	T2	Built
FR-003	Phone normalisation. The system shall normalise Vietnamese mobile numbers (+84..., 84..., missing leading 0, spaces) to 0XXXXXXXX and E.164. Invalid numbers are ignored for survivorship.	Prefixes 03/05/07/08/09, 10 digits.	M	T2	Built
FR-004	Golden profile per vehicle. The system shall merge all records for the same plate into one golden profile. Name and phone are chosen by source trust (survivorship). Alternative phones are retained. Source list and record IDs are kept.	enrichment.sourceTrust (e.g. tasco_core 1.0, vetc_account 0.85, partner_agent 0.5, telesales_csv 0.4).	M	T2	Built
FR-005	Expiry inference with confidence. The system shall infer the current TNDS expiry from available evidence: verified certificate, partner policy record, customer declaration, inspection cycle (+365 days) or tag anniversary. Dates are rolled forward a year at a time until within 60 days of today. The best evidence is picked, and confidence is boosted when independent methods agree within 21 days. All candidates are kept.	enrichment.expiryEvidence, corroboration (+0.15 per agreement, cap 0.95), rollForwardFloorDays 60, usableExpiryConfidence 0.5.	M	T2	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-006	Vehicle category inference. The system shall infer the tariff category (11 car categories) from declared seats, usage, owner type and toll class using a first-hit decision table, and record its confidence and basis.	<code>enrichment.categoryTable</code> ; default <code>car_under6</code> at 0.3.	M	T2	Built
FR-007	Data-quality scoring and issue queue. The system shall compute a DQ score (0–100) per profile and raise issues for missing <code>phone</code> , <code>name</code> , <code>reliable_expiry</code> , <code>vehicle_category</code> and <code>current_insurer</code> , and for <code>conflicting_phone</code> . Bot outcomes add <code>wrong_person</code> and <code>plate_mismatch</code> ; ingestion adds <code>invalid_plate</code> . Data stewards list issues by type and status and resolve them with a written resolution.	<code>enrichment.dataQuality</code> weights 0.5 / 0.35 / 0.15; <code>categoryMinConfidence</code> 0.6.	M	T2	Built
FR-008	Field-level lineage. For any profile, authorised staff shall see the source, confidence and rule behind each derived field (<code>phone</code> , <code>name</code> , <code>category</code> , <code>expiry</code>), the contributing record IDs and sources, and recent ingestion batches.	—	S	T2	Built
FR-009	Steward expiry correction. A data steward (<code>profile:update</code>) shall be able to set expiry date and insurer with mandatory evidence text . The change is audited and the lead is recomputed immediately.	<code>service_levels.stewardCorrectionConfidence</code> (0.9).	M	T2	Built
FR-010	Data-repair loop. Facts captured on the platform shall update the golden profile and survive later source rebuilds when they carry higher confidence: customer-declared expiry, bot-captured "already renewed" (next expiry and competitor note), TASCO issuance (confidence 1.0, method <code>tasco_issued</code>) and consent changes.	Methods that survive rebuilds: <code>customer_declared</code> , <code>voice_bot</code> , <code>tasco_issued</code> ; <code>consentOverrides</code> .	M	T2	Built
FR-011	Incremental rebuild. Ingestion shall rebuild only the profiles whose plates appear in the batch (chunks of 500 plates), then emit <code>profiles.rebuilt</code> so that only the affected leads are re-scored.	Event-driven via the outbox.	M	T2	Built

B. Lead intelligence

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-012	Fact model. The system shall derive a standard fact set for every vehicle and use it consistently across all rule kinds: days to expiry, lapsed days, expiry confidence and method, insurer, tag age, owner type, region, category, seats, usage, vehicle age, reachability channels, consent, premium and engagement metrics.	<code>src/domain/leads.js#factsFor</code>	M	T2	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-013	Journey assignment. Each vehicle shall be placed in the first journey (by priority) whose audience expression is true: <code>lapsed_uninsured</code> (1), <code>new_vehicle</code> (2), <code>renewal</code> (3), <code>conquest</code> (4).	journeys	M	T3 / NB	Built
FR-014	Explainable lead score. The system shall compute a 0–100 score as the weighted sum of five factors (urgency 35, data confidence 15, engagement 20, reachability 15, relationship 15). The score is damped by expiry confidence ($\times(0.6 + 0.4 \times \text{confidence})$) and set to zero when the customer is on DNC. Tiers: hot ≥ 70 , warm ≥ 45 , else nurture. Each factor shows points, maximum and a plain-language reason.	scoring (weights must sum to 100; hot > warm, enforced by the validator)	M	T2	Built
FR-015	Next-best action (NBA). The system shall choose one action per vehicle using a first-hit decision table: <code>suppress</code> (DNC), <code>route_b2b</code> (company), <code>verify_expiry</code> (confidence < 0.5), <code>welcome_new_vehicle</code> , <code>nurture</code> (> 45 days), <code>urgent_recovery</code> (lapsed), <code>voice_bot</code> (hot + call consent + phone), <code>digital_reminder</code> , <code>sms_reminder</code> , <code>default_enrich</code> . Each action has a label and a reason.	nba	M	T2	Built
FR-016	Personalised benefits per lead. The system shall rank eligible benefits by relevance and attach the top 3 (with a "why" text) to each lead.	benefits (maxShown 3)	M	T4	Built
FR-017	Prioritised lead queue. Staff with <code>leads:read</code> shall list leads filtered by tier, journey, action, region, maximum days to expiry and minimum score, sorted by score or expiry, with pagination (≤ 500 per page). Users without <code>dashboard:read</code> and with a specific region see only their region.	ABAC <code>regional_data</code>	M	T2	Built
FR-018	Recompute. Leads shall be recomputed (a) on demand by a campaign manager (<code>leads:recompute</code>), (b) on events (<code>profiles.rebuilt</code> , <code>lead.recompute_requested</code> , <code>policy.issued</code>) and (c) by a scheduled job (<code>npm run job recompute</code>). If the journey or tier changes, scheduled touchpoints are re-planned.	—	M	T2	Built
FR-019	Insured suppression. A vehicle with an active TASCO TND policy beyond today shall have no journey and the NBA <code>insured</code> . All its scheduled touchpoints are cancelled at execution time.	—	M	T3	Built

C. Journeys and messaging

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-020	Journey catalogue. The system shall support configurable journeys, each with objective (<code>retention</code> , <code>new_business</code> , <code>cross_sell</code>), audience, anchor (<code>expiry</code> , <code>tagActivatedAt</code> , <code>today</code>) and timed steps (<code>offset</code> , <code>channels</code> , <code>marketing flag</code> , <code>template</code> , <code>optional tier gate</code>). It ships with: <code>lapsed_uninsured</code> , <code>new_vehicle</code> , <code>renewal</code> , <code>conquest</code> and post-purchase <code>cross_sell</code> .	journeys (validator: unique IDs, valid anchor, channels required)	M	T3 / NB	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-021	Touchpoint planning. For each lead the system shall plan touchpoints from the journey's steps relative to the anchor date. Steps more than <code>catchUpDays</code> (2) in the past are skipped, and tier-gated steps are omitted for other tiers. Touchpoint IDs are deterministic (<code>profile:journey:step:date</code>) so planning can safely be repeated.	<code>journeys.catchUpDays</code> , <code>onlyTiers</code>	M	T3	Built
FR-022	Execute due touchpoints. On a schedule or on demand (<code>journeys:run</code>), the system shall execute every touchpoint due up to a given date. The run pages through all due touchpoints. For each one it tries the step's channels in order and records the outcome (<code>done</code> , <code>skipped</code> with reasons, <code>cancelled</code>) and per-channel and per-journey totals. A touchpoint blocked only by the contact window is deferred (stays scheduled for the next in-window run). Simulated run dates are honoured only in demo mode. A <code>voice_bot</code> step starts an automated call; a <code>telesales</code> step creates a handoff task.	—	M	T3	Built
FR-023	Contact policy gate. Every outbound contact shall pass a policy check before sending: channel reachable; not DNC; consent present (<code>marketing</code> for marketing; <code>call</code> for voice bot and telesales); marketing only between 08:00 and 20:00 ICT ; at most 1 marketing contact per day and 3 per week ; at most 2 call attempts per week . Service messages bypass the frequency caps.	<code>contact_policy</code>	M	PL	Built (journeys, triggers and voice campaigns)
FR-024	Copy guard. Customer copy shall never contain banned discount or rebate phrases. The check runs (a) when content, benefit or voice-bot rule sets are saved, which blocks the draft, and (b) at send time, where the message is recorded as <code>blocked</code> and not sent. Matching ignores diacritics, extra spaces and punctuation (e.g. "giảm giá", "cash back").	<code>copy_guard</code>	M	T4 / PL	Built
FR-025	Templated bilingual messages. Messages shall be rendered from templates per key with Vietnamese for customers and English for staff review. Placeholders: <code>{{plate}}</code> , <code>{{expiry}}</code> (dd/mm/yyyy), <code>{{days}}</code> , <code>{{premium}}</code> , <code>{{benefit}}</code> (top benefit), <code>{{link}}</code> (signed renewal deep link), <code>{{certNo}}</code> .	<code>content.messages</code> (13 templates)	M	T3	Built
FR-026	Moments of truth (ecosystem triggers). The system shall accept VETC events <code>vetc.tag_activated</code> , <code>vetc.inspection_booked</code> , <code>vetc.wallet_topped_up</code> and <code>vetc.long_trip_started</code> . Matching triggers either enrol the vehicle in a journey or send a templated message immediately, subject to the trigger condition and the contact policy.	<code>triggers</code>	S	NB / T3	Built (API and simulator); live VETC event feed planned
FR-027	Purchase confirmation. After issuance the system shall send a service message (<code>purchase_confirmation</code>) with certificate number, plate, expiry and a verification link, using the first available channel of app push or Zalo.	<code>content.messages</code>	M	T3	Built
FR-028	Post-purchase cross-sell. After a TND5-only purchase by a customer with marketing consent, the system shall schedule the <code>cross_sell</code> step (+1 day) via push or Zalo, highlighting cover the customer does not yet hold.	<code>journeys.crossSell</code>	S	NB / T4	Built
FR-029	Touchpoint schedule view. Staff shall list touchpoints by profile and status (<code>scheduled</code> , <code>done</code> , <code>skipped</code> , <code>cancelled</code>).	—	S	T3	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-030	Journey integrity. A touchpoint shall be cancelled when the profile is missing or anonymised, when the vehicle is already insured with TASCO, or when the lead's journey has changed since planning.	—	M	T3	Built

D. AI voice bot

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-031	Disclosure. Every call shall open by saying that this is VETC's automated assistant , that the call is recorded, and that VETC never asks for OTPs or payment by phone. It then asks for the licence plate.	content.voicebot → disclosure, lines.intro	M	T1	Built
FR-032	Plate-first verification. The bot shall extract the plate from the customer's speech (Vietnamese and English spoken digits supported) and compare it with the record. The bot never reads the plate out. No plate recognised → retry (maximum 3 attempts) → outcome <i>unverified</i> . Mismatch → polite end with outcome <i>plate_mismatch</i> , and a DQ issue is raised.	content.voicebot.maxPlateAttempts	M	T1	Built
FR-033	Expiry confirmation. After verification the bot shall state the expiry on file using the masked plate. Three variants: known (confidence ≥ 0.5), lapsed or unknown ("which month does it end?"). The customer's answer is captured as a signal.	Lines expiryKnown, expiryLapsed, expiryUnknown	M	T1 / T2	Built
FR-034	Intent recognition. The bot shall classify each utterance into one of 12 intents: <i>opt_out</i> , <i>scam_concern</i> , <i>human</i> , <i>already_renewed</i> , <i>price</i> , <i>callback_later</i> , <i>send_link</i> , <i>buy_now</i> , <i>wrong_person</i> , <i>benefits</i> , <i>yes</i> , <i>no</i> . It uses accent-insensitive keyword matching, and an LLM or NLU classifier can be plugged in through the same interface.	content.voicebot.intents	M	T1	Built (keyword); vendor NLU planned
FR-035	Price honesty and value pivot. When asked about price, the bot shall state that TND\$ premiums are set by regulation and identical at every insurer, give the customer's premium including VAT, and pivot to service value. It never offers a discount.	Line price; copy_guard	M	T1 / T4	Built
FR-036	Call outcomes and write-back. Each call shall end in exactly one outcome, with an action for each: • <i>hot_handoff</i> → telesales handoff • <i>link_sent</i> → renewal link sent as a service message via app, Zalo or SMS • <i>already_renewed</i> → next expiry = previous + 365 days at confidence 0.6 (<i>service_levels</i>); insurer OTHER; competitor note • <i>opted_out</i> → DNC and call consent withdrawn, audited • <i>wrong_person</i> / <i>plate_mismatch</i> → DQ issue • <i>callback_later</i>, <i>not_interested</i>, <i>unverified</i> → logged	service_levels	M	T1 / T2	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-037	Interactive console. Authorised staff (voice:operate) shall be able to start a session for a customer (refused if DNC), send customer utterances as text and view the bilingual transcript. Used for rehearsal, QA and supervised calls.	—	S	T1	Built
FR-038	Automated call campaign. A campaign manager (journeys:run) shall run a bot campaign over the top leads of a tier (hot or warm, up to 200 calls). Company-owned vehicles are excluded, and each call must pass the full contact policy (call and marketing consent, not DNC, 08:00–20:00 window, weekly call cap). A summary of outcomes and skipped reasons is returned.	contact_policy	M	T1	Built
FR-039	Voice governance KPIs. The system shall report call volume, outcomes, plate-verification failure rate, opt-out rate and the disclosure statement.	—	S	T1 / PL	Built

E. Telesales

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-040	Structured handoff. A hot bot call shall create a handoff holding only what the agent needs: plate, plate-verified flag, name, masked phone, journey, expiry, days to expiry, premium, score, outcome, price-asked and trust-concern flags, and talking points . Talking points include: call from the official hotline, never take payment by phone, the regulated-price message, the top benefits, and closing by sending the one-tap link.	Data minimisation	M	T1	Built
FR-041	Work queue. Agents shall see open and assigned handoffs filtered by status, "mine" and region, with pagination. Agents see only handoffs that are unassigned or assigned to them.	ABAC agent_own_handoffs, region filter	M	T1	Built
FR-042	Handoff lifecycle. Handoffs shall follow open → claimed / callback / won / lost, claimed → won / lost / callback / open and callback → claimed / won / lost. won and lost are terminal. Claiming assigns the handoff to the agent. Notes are appended with author and time. Optimistic version control prevents lost updates.	—	M	T1	Built
FR-043	Supervisor assignment. A supervisor (handoff:assign) shall be able to assign or reassign any handoff.	RBAC	M	T1	Built
FR-044	Journey escalation to telesales. A journey telesales step shall create a direct handoff (reason journey_escalation) with talking points for customers who have not responded to digital reminders.	journeys (onlyTiers: ["hot"])	M	T3	Built

F. Sales and issuance

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-045	Product catalogue. The system shall expose active products (TNDS_CAR, TNDS_MOTORBIKE, MOTOR_PD, PA_SEAT) with their eligible channels, and bundles (SAFE_DRIVE, FULL_MOTOR).	products	M	NB	Built
FR-046	Quotation. For a profile the system shall quote 1–5 product lines for a channel. It checks that each product is sold on that channel. Cover starts the day after current expiry, or today if lapsed or unknown. Each line shows net premium, VAT, total, start and end dates and breakdown. The quote attaches benefits, detects a matching bundle and expires after 24 hours . Telesales and supervisors send a staff quote to the customer's VETC app or Zalo (POST /api/quotes/:id/send), where the customer sees it (GET /api/customer/quotes).	products, rating kinds, service_levels.quoteTtlHours	M	T3 / NB	Built
FR-047	Pay and issue (idempotent). Only the customer (in the VETC app) or a partner can pay; staff never take payment . Purchase requires an Idempotency-Key; a retry returns the original result (idempotentReplay: true), and reusing the key of a failed attempt returns HTTP 409. The quote is claimed atomically (status paying), so concurrent purchases cannot double-charge. Flow: debit VETC wallet (partner sales: partner-collected) → issue each line through TASCO core → store policy with certificate number and verification URL → mark quote converted.	—	M	T3	Built (sandbox ports); real wallet and core integration planned
FR-048	Failure compensation. If payment fails, the order is set to payment_failed. If issuance fails after payment, any issued lines are cancelled and the payment is refunded in full through the circuit breaker; the order is set to issuance_failed_refunded, or compensation_failed if the compensation itself fails. All outcomes are audited.	—	M	PL	Built
FR-049	Golden record update on issuance. A TNDS issuance shall set the profile's expiry to the policy end date, method tasco_issued, confidence 1.0, insurer TASCO and verified = true.	—	M	T2	Built
FR-050	Public certificate verification. Anyone holding a certificate number (QR) shall be able to verify validity, status, product, masked plate, insurer and period, with no PII. The endpoint is rate-limited.	—	M	T4 / PL	Built (API and page /verify/:certNo)
FR-051	Policy listing. Staff (policy:read) shall list policies by profile and product.	—	M	T3	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-052	Reconciliation. A job shall check that every completed order has a payment reference and that each referenced policy exists, and flag orders stuck in <code>pending_payment</code> for more than 1 hour, <code>payment_failed</code> orders and <code>compensation_failed</code> orders.	—	M	PL	Built

G. Products and rating

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-053	Regulated tariff rating (TNDIS). Annual premium by category, excluding VAT, pro-rata by days over 365. Term 1–3 years. VAT 10%. Car and motorbike tariffs per Decree 67/2023/ND-CP, to be confirmed by TASCO underwriting.	<code>tariff.tnds_car</code> , <code>tariff.tnds_motorbike</code> (validator: VAT 0–0.2, positive integer premiums)	M	T3	Built
FR-054	Physical damage rating (MOTOR_PD). Premium = sum insured × rate (first-hit by usage and vehicle age) × (1 – deductible relief). Minimum VND 1,000,000. Sum insured above VND 5 bn is referred to an underwriter. Deductible options 0 / 500,000 / 1,000,000. A recorded vehicle inspection (<code>POST /api/quotes/:id/inspection</code> , passed + evidence) is required before payment; one-tap purchase is blocked until then.	<code>rating.motor_pd</code> (ILLUSTRATIVE rates)	S	NB	Built
FR-055	Personal accident per seat (PA_SEAT). Premium = seats (1–60) × sum insured per seat (10 / 20 / 50 / 100 M) × rate per seat. VAT-exempt.	<code>rating.pa_seat</code> (ILLUSTRATIVE)	S	NB	Built
FR-056	No-discount pricing. Quotes for price-regulated products shall carry no discount field and shall state "Premium fixed by regulation — identical at every insurer".	<code>products.priceRegulated</code>	M	T4 / PL	Built
FR-057	Products by configuration. A new product shall be added by configuration: code, rating method (<code>tariff_table</code> , <code>rate_on_sum_insured</code> , <code>per_seat</code>), rate rule kind, channels and status. The validator rejects unknown methods, duplicate codes and bundles that reference unknown products.	<code>products</code>	S	PL	Built

H. Value and benefits

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-058	Benefits catalogue with legal gating. Benefits shall have type (service, loyalty, convenience, cover upgrade), provider, bilingual title and description, eligibility, relevance and a "why" expression. Only <code>legalStatus = approved</code> items are shown to customers. Items with <code>available: false</code> are roadmap items shown to staff only. Staff see all items, with pending and roadmap ones flagged.	<code>benefits</code>	M	T4	Built
FR-059	Relevance and explanation. Benefits shall be ranked by a per-customer relevance score (0–1) with a plain-language reason (for example "drives 1,850 km on highways per quarter").	<code>benefits</code>	M	T4	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-060	Loyalty points and referral. Non-cash VETC points for renewal or data verification, and a referral programme (200 points per referral, maximum 5 per month). Both are disabled pending legal approval.	benefits.loyalty_points (pending), referral (enabled: false)	C	T4	Planned (gated)
FR-061	Auto-renew opt-in. The customer opts in once, is reminded 7 days before expiry and is renewed from the VETC wallet only after confirming .	benefits.auto_renew (available: false)	S	T3 / T4	Planned (roadmap; staff view only)
FR-062	Inspection (đăng kiểm) reminder and booking. Track the inspection date, remind the customer and help book a slot. Use vetc.inspection_booked as a trigger to check TND5 validity.	benefits.inspection_assist, triggers.inspection_booked	S	T4 / NB	Partial (trigger built; booking integration planned)

I. Partners and fleet

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-063	Partner onboarding and status. A partner manager shall onboard partners of type bank, showroom, agent, fleet or inspection_center with a region, and activate or suspend them. A suspended partner's keys stop authenticating.	—	M	NB	Built
FR-064	Partner API keys. The system shall issue API keys (prefix tpk_) that are displayed once and stored only as SHA-256 hashes, with scopes quote, purchase, policies:read. Keys can be revoked.	—	M	NB / PL	Built
FR-065	Partner quote by plate (new business). A partner shall quote by plate, holder name, phone, seats, usage, current expiry and marketing consent. If the plate is unknown, or new facts are supplied, a partner source record is ingested so that the vehicle joins the base. Owner type (individual or company) may be supplied. Quotes are tagged with the partner and channel partner_api.	products (channel eligibility); enrichment.sourceTrust.partner_*	M	NB	Built (residual risk E-11)
FR-066	Partner bind. A partner shall bind only its own quotes, with an Idempotency-Key. Partner sales are partner-collected: no VETC wallet debit, payment reference prefixed PARTNER-, premium remitted to TASCO under the partner agreement and commission statement.	—	M	NB	Built
FR-067	Partner self-service reporting. A partner shall list the policies it sold and its own commission statement for a date range.	—	M	NB	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-068	Commission calculation. Commission per policy line = net premium × rate. The rate comes from a first-hit table by product and partner type and is capped at the statutory cap . The validator rejects any rule above the cap.	commission (TNDS 5%; PD 10% showroom / 8% other; PA 15%; caps to be confirmed by TASCO finance and legal)	M	NB / PL	Built
FR-069	Partner statement (staff). A partner manager shall view any partner's commission statement: completed orders, lines and total.	—	M	NB	Built
FR-070	Fleet / B2B. Company-owned vehicles shall be routed to the B2B team (NBA route_b2b), excluded from consumer journeys and automated bot campaigns, and offered a fleet renewal dashboard and consolidated VAT invoice.	nba.fleet, journeys (audiences), benefits.fleet_dashboard	S	NB	Partial (routing and journey exclusion built; fleet dashboard and invoice planned)

J. Claims FNOL

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-071	Report an accident (FNOL). A customer shall report an incident against one of their own policies: incident date (must be within the policy period), description, location and photo count. The claim ID has the form CL-XXXXXXX. An acknowledgement SLA due time (4 hours) is set and an acknowledgement message (claim_received) is sent to the customer.	service_levels.claimAckSLaHours	S	T4	Built (API and app; photo count only, photo upload planned)
FR-072	Claims queue and workflow. Claims handlers shall list claims by status and advance them through submitted → acknowledged → assessor_assigned → under_assessment → approved → paid, with rejected allowed from early states. Every transition is recorded in history and the audit trail.	—	S	T4	Built
FR-073	My claims. A customer shall see the status of their own claims.	—	S	T4	Built

K. Customer self-service and privacy

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-074	Customer session. The customer app (VETC app web view or Zalo mini app) shall exchange a signed renewal link (HMAC, no enumerable IDs) or a VETC SSO token for a 1-hour customer token scoped to one vehicle. Signed links expire after LINK_TTL_DAYS (default 30). Login is rate-limited.	ABAC customer_self	M	T3	Built (signed link); VETC SSO planned

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-075	Cover status home. The customer shall see vehicle, cover (expiry, confidence, insurer, verified, days to expiry, and a "please confirm" prompt when confidence < 0.75), regulated premium, TASCO policies with certificate links, top approved benefits and current consent.	benefits	M	T3 / T4	Built
FR-076	Confirm my expiry. The customer shall declare the current expiry date and insurer. This updates the golden record (customer_declared, confidence 0.75) and triggers recompute.	service_levels.customerDeclaredConfidence	M	T2	Built
FR-077	One-tap renew with add-ons. The customer shall get a quote for their vehicle (TNDS, optionally with PA_SEAT or MOTOR_PD, term 1–3 years) and pay from the VETC wallet with an Idempotency-Key; quotes sent by telesales appear for confirmation. MOTOR_PD requires a recorded inspection first (FR-054). They receive the e-certificate with a QR immediately.	Channel vetc_app	M	T3	Built (API and customer app)
FR-078	Consent centre. The customer shall switch marketing and call consent on or off. Changes are audited, survive data rebuilds and trigger recompute.	contact_policy.consentRequired	M	PL	Built
FR-079	Download my data. The customer shall download all data held about their vehicle: profile, lead, policies, quotes, orders, claims, telesales tasks, messages, source records and voice sessions.	PDP right of access	M	PL	Built
FR-080	Staff DSAR handling. A compliance officer (dsar:manage) shall export a data subject's data and erase it by anonymisation. Erasure is refused while an active policy is in force (legal retention). Messages, transcripts, voice signals, handoffs and claim text are scrubbed, source PII is nulled and the subject's sessions are invalidated.	retention	M	PL	Built
FR-081	PII masking by permission. Customer 360 shall mask name and phone unless the viewer holds profile:read_pii. Every profile view is audited with a piiVisible flag.	rbac	M	PL	Built

L. Rules governance

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-082	Versioned rule registry. All rule kinds shall be stored as versions (kind@n) with status draft, pending_approval, active, retired or rejected, plus checksum, author, approver and timestamps. Staff with rules:read list versions and view payloads.	All kinds	M	PL	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-083	Draft with validation. A rule author shall create a draft only if it passes validation: every embedded JSON Logic expression and decision table, the kind-specific checks (FR-014, FR-020, FR-053, FR-057, FR-068, contact window) and the copy guard for content and benefit kinds.	src/rules/validators.js	M	PL	Built
FR-084	Validate without saving. Authors shall be able to check a payload and see the list of errors.	—	S	PL	Built
FR-085	Simulate. Staff shall run a candidate <code>scoring</code> , <code>nba</code> , <code>journeys</code> or <code>benefits</code> payload against a real customer and compare the current and candidate evaluations (journey, score, tier, reasons, NBA, benefits) before submitting.	—	S	T2 / PL	Built (4 kinds)
FR-086	Maker-checker approval. Only the author can submit a draft. Only a different user with <code>rules:approve</code> can approve or reject it, with a comment. Restricted kinds (<code>abac</code> , <code>copy_guard</code> , <code>contact_policy</code> , <code>commission</code> , <code>retention</code>) can be approved only by a <code>compliance_officer</code> . Approval retires the previous active version and activates the new one in one transaction; the database enforces one active version per kind. All replicas pick up the change within about 15 seconds (cache TTL and <code>rules.activated</code> event).	rbac.restrictedRuleKinds	M	PL	Built
FR-087	Rollback. Rolling back shall create a new draft copied from an earlier version, which still requires approval.	—	M	PL	Built
FR-088	Default rule seeding. On first start the system shall seed each kind from <code>config/rules/*.json</code> (copy guard first) as version 1, validated, with an audit entry.	—	M	PL	Built

M. Identity and access

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-089	Staff login with logout. Username and password login. After 5 failures (password or MFA code) the account is locked for 15 minutes (configurable). Timing is equalised for unknown users. Every attempt is audited.	Config LOCKOUT_*	M	PL	Built
FR-090	MFA (TOTP). A second factor is required for enrolled users and mandatory for <code>admin</code> , <code>rule_approver</code> , <code>compliance_officer</code> and <code>data_steward</code> . Users in these roles self-enrol at first sign-in (QR code shown only to them; administrators never see seeds). Codes cannot be reused.	Config MFA_REQUIRED_ROLES	M	PL	Built
FR-091	Sessions. Short-lived access tokens (default 30 minutes) carrying roles, region and authentication method. Logout revokes the token; a password change or a role, status or region change revokes all existing tokens.	—	M	PL	Partial (logout revocation per replica, E-15)
FR-092	RBAC. 15 roles mapped to permissions, deny by default. Segregation of duties: forbidden role pairs (e.g. <code>admin</code> + <code>rule_approver</code> , <code>rule_author</code> + <code>rule_approver</code>) are defined in <code>rbac.json</code> ; the platform admin cannot author or approve rules or see customer PII.	rbac	M	PL	Built
FR-093	ABAC. Attribute policies evaluated after RBAC: agents see only their own or unassigned handoffs; telesales see profiles in their region; customers see only their own vehicle. Adding an unrelated role never widens access.	abac	M	PL	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-094	User administration. An admin shall create users (password policy, roles, region), change roles, region or status, list users, and unlock an account or force MFA re-enrolment (POST /api/users/:id/reset). New users must change their password at first sign-in. An admin cannot reset their own account or change their own roles or status.	—	M	PL	Built
FR-095	Partner and customer authentication. Partner calls authenticate with X-Api-Key and receive role partner_api. Customer tokens (audience customer) are scoped to one profile.	—	M	PL	Built
FR-096	Password management. Users shall change their password after re-entering the current one. Policy: 12–128 characters (OWASP ASVS V2.1).	—	M	PL	Built

N. Audit

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-097	Audit every state change. The system shall record actor, action, entity and details for authentication, user, rule, data, lead, journey, voice, handoff, quote, order, partner, claim, consent, DSAR, DQ and job events, including profile views with the PII-visibility flag.	—	M	PL	Built
FR-098	Audit search. Auditors and compliance staff (audit:read) shall search by entity, actor and action with pagination.	—	M	PL	Built
FR-099	Tamper evidence. Audit entries shall be hash-chained (each hash covers the previous one, using canonical JSON). A verify function reports ok or the index where the chain breaks.	—	M	PL	Built

O. Operations, reporting and user interfaces

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-100	Growth overview dashboard. Base size, expiring in 30 days, lapsed, usable-data profiles; leads by tier, journey and action; DQ issues by type; messages by channel and status; bot outcomes; handoffs; orders and premium by journey and channel; active policies by product; claims; economics (bot cost, telesales-equivalent cost, saving, messaging cost, cost per order).	costs	M	PL	Built (scaling note E-16)
FR-101	Adoption dashboard. Active users, logins, failed logins, handoff actions, actions by type, and adoption targets (weekly active telesales $\geq 90\%$, ≤ 3 clicks to renew, handoff first contact within 2 business hours, rule-change lead time ≤ 1 day).	—	S	PL	Built
FR-102	Governance dashboard. Voice-bot governance (FR-039), rule-set counts by status and audit-chain verification.	—	S	PL	Built

ID	Requirement	Business rules / rule kind	Pri	Track	Status
FR-103	Operational status. Store type, circuit-breaker state for each integration (wallet, core, telephony, push, ZNS, SMS), active rule versions with checksums, event backlog by status and audit entry count.	—	M	PL	Built
FR-104	Jobs. Support engineers (<code>ops:run_jobs</code>) shall run <code>reconciliation</code> , <code>retention</code> and <code>relay</code> (outbox) jobs and view job history. The same jobs run on a schedule via the CLI.	—	M	PL	Built
FR-105	Retention execution. Apply per-entity retention: source records deleted after 365 days, voice sessions after 180 days, profiles anonymised after 1,825 days of inactivity, and messages, orders, certificates and audit archived after the stated periods. Legal hold overrides deletion.	<code>retention</code>	M	PL	Partial (E-09)
FR-106	Health, metrics and API description. Liveness and readiness probes (readiness checks the DB), Prometheus metrics and an OpenAPI 3 document generated from the route table.	—	M	PL	Built
FR-107	Demo and UAT sandbox. Seed demo users (one per role), 5 demo partners and a synthetic VETC base that reproduces the brief's data problems (about 1 in 10 verified). Demo mode is opt-in (<code>DEMO_MODE</code> , default off); the seed job and the demo TOTP helper require it. UAT-only switches (<code>DEMO_PASSWORD</code> , <code>DEMO_TOTP_SEED</code> , <code>DEMO_ACCOUNT_SYNC</code>) manage demo accounts. Production refuses to start without <code>DATABASE_URL</code> .	—	S	PL	Built
FR-108	Staff web console. Login with MFA (no role shortcut buttons); role-based home dashboard; leads queue with explainable score and NBA; Customer 360 (lineage, journey timeline, messages, policies, benefits); voice bot console; telesales inbox; journeys (run due, ecosystem event simulator); rules studio (draft, validate, simulate, submit, approve); partners (onboard, keys, statement); claims queue; data quality; audit (search, verify); users; operations; contextual help drawer.	—	M	All	Built (public/)
FR-109	Customer mobile app (/app/). Cover status card, confirm expiry, one-tap renew with add-ons, VETC wallet pay, e-certificate with QR, benefits, consent centre, report accident, download my data. Public certificate page <code>/verify/:certNo</code> .	—	M	T3 / T4	Built (public/app/, public/verify.html)
FR-110	Localisation and themes. Vietnamese and English toggle in the staff console and customer app (Vietnamese is the customer default), light and dark themes, and dates in dd/mm/yyyy for Vietnamese.	—	M	PL	Built

Appendix A. Rule-kind index

Rule kind	File	Governs FRs
enrichment	<code>config/rules/enrichment.json</code>	FR-004 – FR-007
scoring	<code>scoring.json</code>	FR-014
nba	<code>nba.json</code>	FR-015, FR-070
journeys	<code>journeys.json</code>	FR-013, FR-020, FR-021, FR-028, FR-044, FR-070
triggers	<code>triggers.json</code>	FR-026, FR-062

Rule kind	File	Governs FRs
contact_policy	contact_policy.json	FR-023, FR-038, FR-078
copy_guard	copy_guard.json	FR-024, FR-035, FR-056
content.messages	content.messages.json	FR-025, FR-027, FR-046, FR-071
content.voicebot	content.voicebot.json	FR-031 – FR-035
benefits	benefits.json	FR-016, FR-058 – FR-062
products	products.json	FR-045, FR-046, FR-057
tariff.tnds_car, tariff.tnds_motorbike	tariff.*.json	FR-053
rating.motor_pd, rating.pa_seat	rating.*.json	FR-054, FR-055
commission	commission.json	FR-068
referral	referral.json	FR-060
retention	retention.json	FR-080, FR-105
costs	costs.json	FR-100
service_levels	service_levels.json	FR-009, FR-036, FR-046, FR-071, FR-076
abac	abac.json	FR-017, FR-041, FR-093
rbac (security config)	config/security/rbac.json	FR-086, FR-092

Appendix B. Out of scope (this release)

- Claims adjudication, reserving and payment. These stay in TASCO core; the platform is the digital front door (FR-071 – FR-073).
- Underwriting of MOTOR_PD above VND 5 bn sum insured (referred to an underwriter).
- Pricing changes to regulated TNDs premiums. These are regulatory and are encoded, not decided, by the platform.
- Cash or premium-reducing incentives of any kind (prohibited).